HOTEL BOOKING

Machine Learning Assessment

From messy booking records to two operational decisions

Tasks 1–8 | Executed notebooks | Reproducible outputs

HMS DATA TEAM

2026

PROJECT SETUP

Project scope and analytical questions

119,390
raw booking rows

33
source columns

2015–2017
arrival period

2
hotel types

Question 1

Which bookings are likely to cancel?

Operations can prioritize confirmation reminders within a fixed review capacity.

Question 2

Which quoted ADR values need review?

Revenue management can compare the quote with a booking-time model expectation.

TASK 1

Data quality rules preserve auditability

119,209

rows after valid removals

37.1%

correct cancellation rate

1,206

No-Show rows retained as cancellations

Rows removed

181 records with no guests or a negative ADR

Rows retained with flags

Exact duplicates, zero-night stays, extreme ADR and large groups

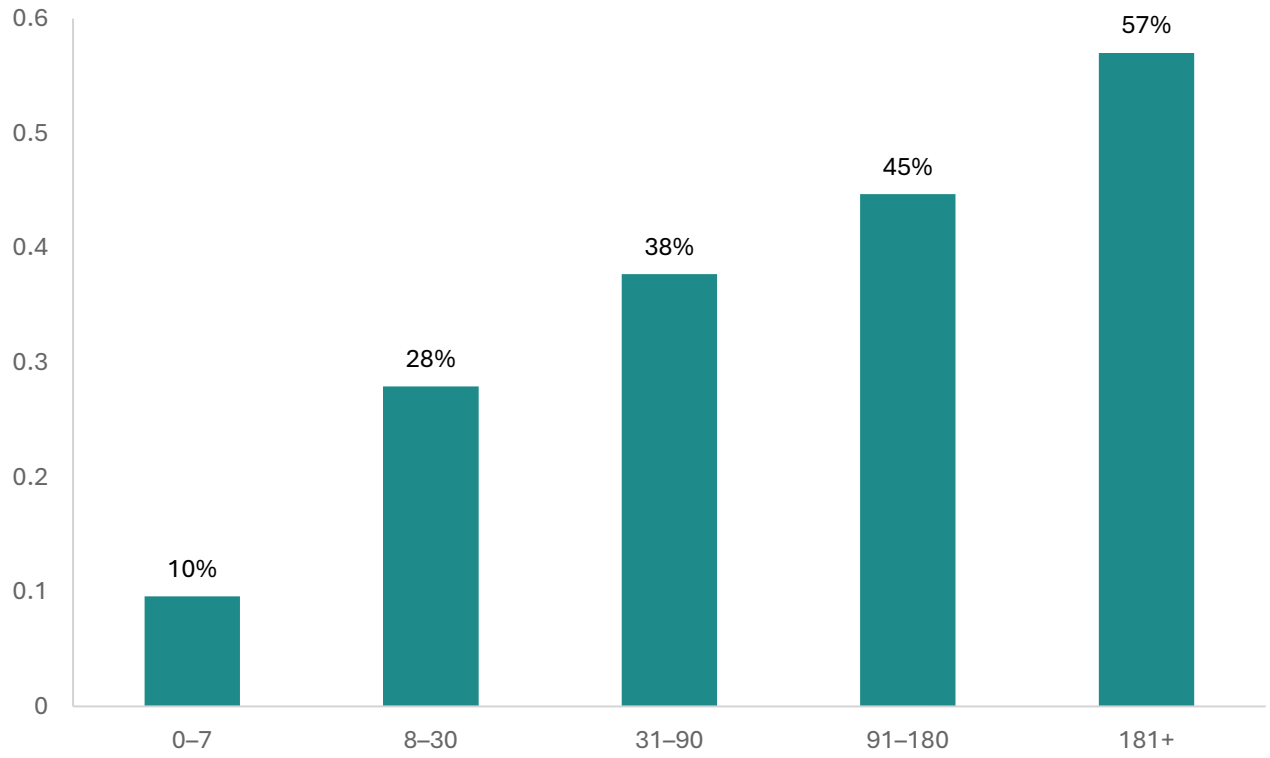
Target correction

Canceled and No-Show both remain cancellation outcomes

The dataset has no booking ID, so identical rows cannot be proven duplicates.

TASK 2

Cancellation risk rises with lead time



Five tested findings

Long lead time is associated with higher cancellation

No special requests signals higher risk

City Hotel cancellation exceeds Resort Hotel

Summer ADR exceeds winter ADR

Non-refundable bookings require policy audit

Tests use chi-square or Mann–Whitney methods, with confidence intervals for rate differences.

TASK 2

Non-refundable bookings reveal a process issue

99.4%

cancellation rate for Non Refund

28.4%

cancellation rate for No Deposit

14,587

Non Refund bookings reviewed

Interpretation

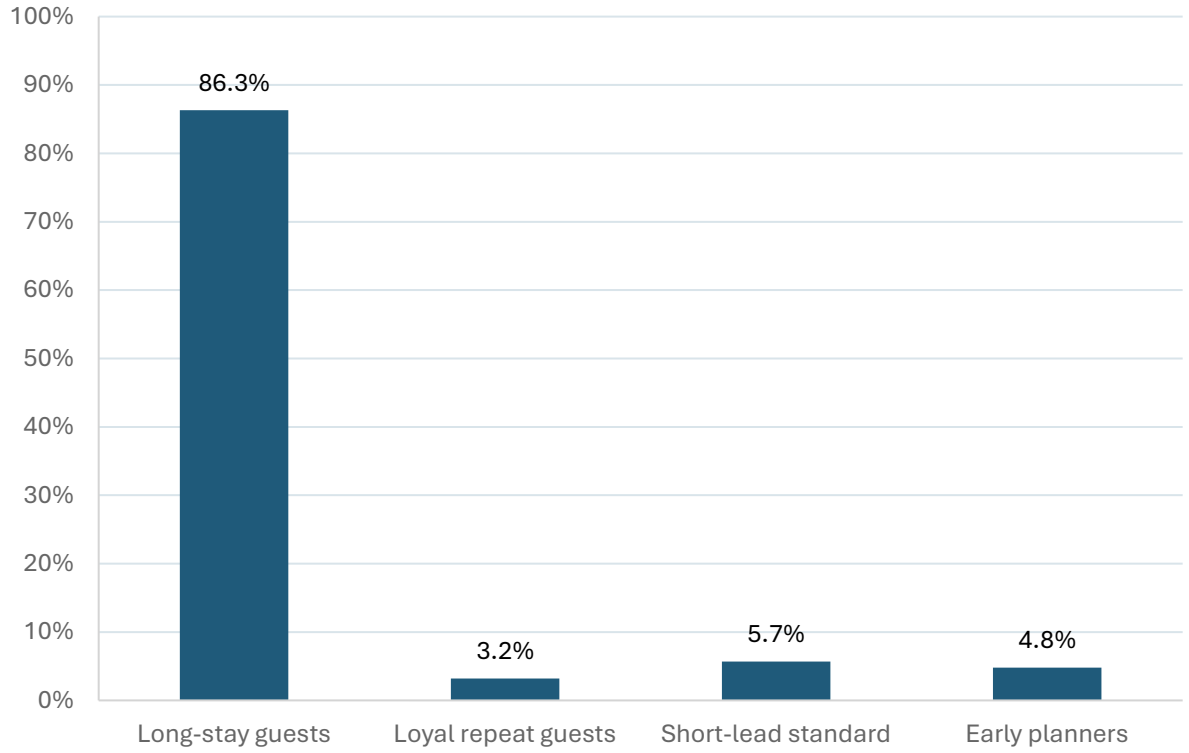
Deposit type may reflect channel rules, group contracts or records created after cancellation risk became known.

Recommended response

Audit the booking channel and contract workflow before changing deposit policy. Test any intervention on a limited sample.

TASK 3

Four segments support different operating actions



Actions

Long-stay guests
Review revenue exposure and extension offers

Loyal repeat guests
Protect benefits and avoid unnecessary friction

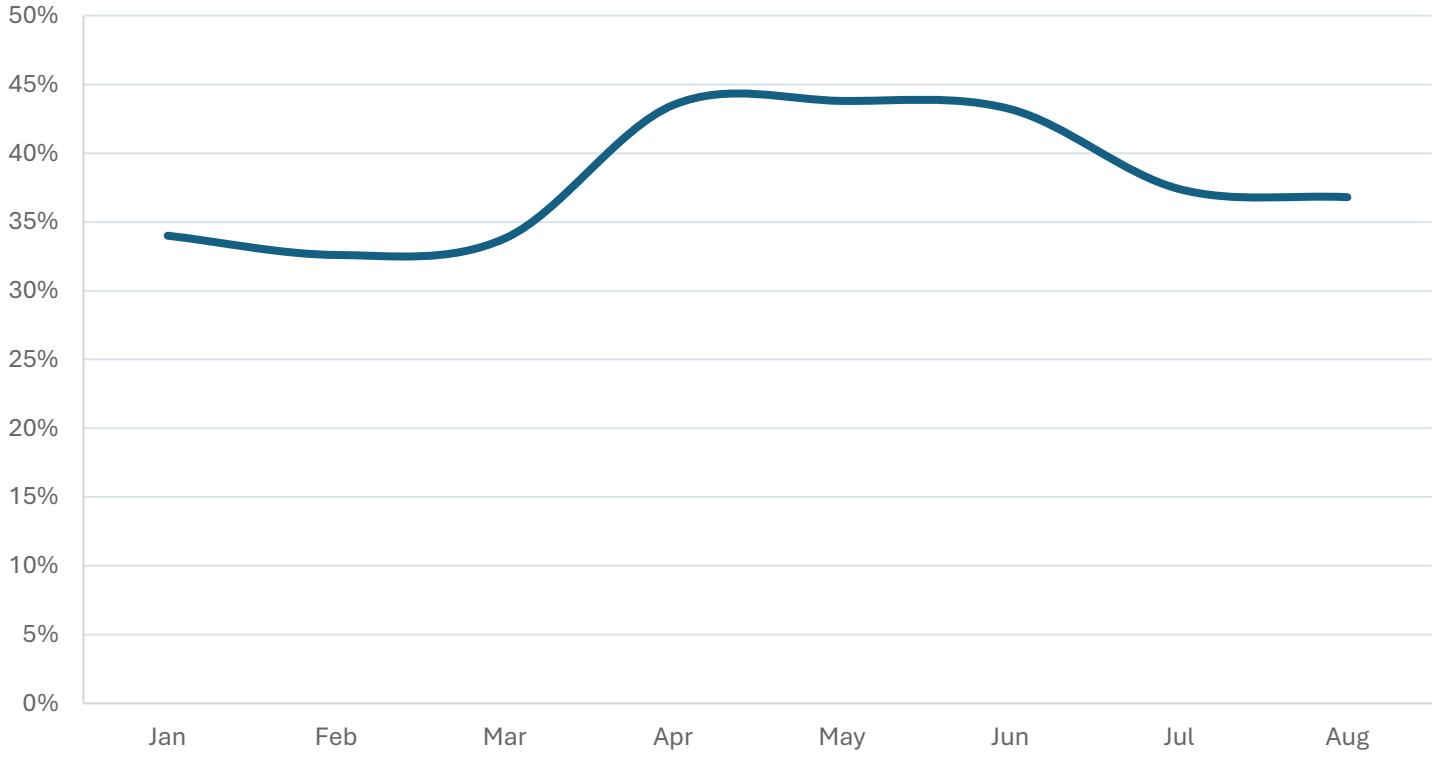
Short-lead standard stays
Use fast confirmation and availability pricing

Early planners
Schedule reminders well before arrival

K=4 balances interpretability with separation. K=2 had the highest silhouette score.

TASK 4

Monthly metrics separate demand from capacity



Metric definitions

Cancellation rate
Canceled and No-Show divided by bookings

Weighted ADR
Completed estimated revenue divided by completed nights

Nights sold
Check-Out nights only

No-Show rate
No-Show divided by Check-Out plus No-Show arrivals

Occupancy is not reported because the source has no room-capacity history.

TASK 5

Booking-time features prevent target leakage

14
engineered features

58,723
training rows

19,867
validation rows

40,619
test rows

Used at booking creation

Lead time, stay pattern, guest count, hotel, market segment, requested room, deposit type and prior history

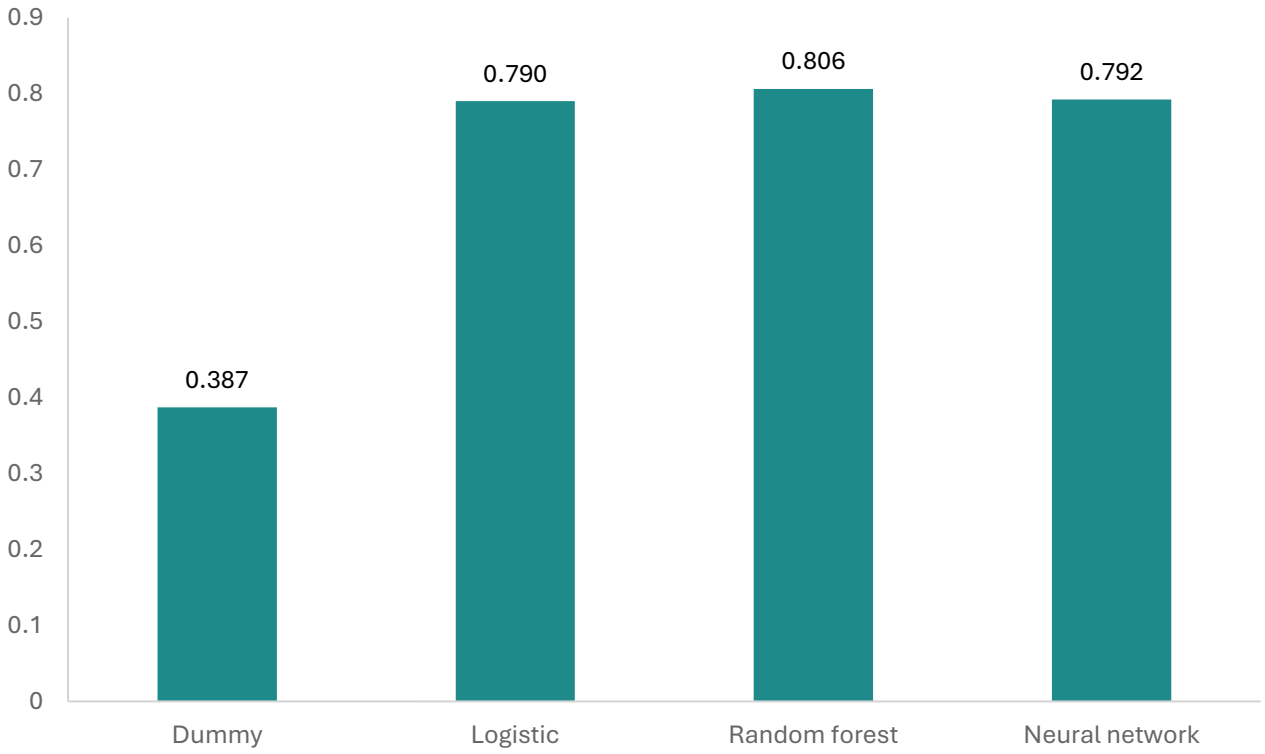
Excluded from modelling

Reservation status, status date, assigned room, booking changes, waiting days and all target-derived fields

Time split: train before Sep 2016, validate Sep–Dec 2016, test Jan–Aug 2017.

TASK 6

Cancellation ranking reaches 0.790 PR-AUC



Selected model

Logistic regression

Validation selected the model before the test period. The top validation quartile set the operating threshold at 0.609.

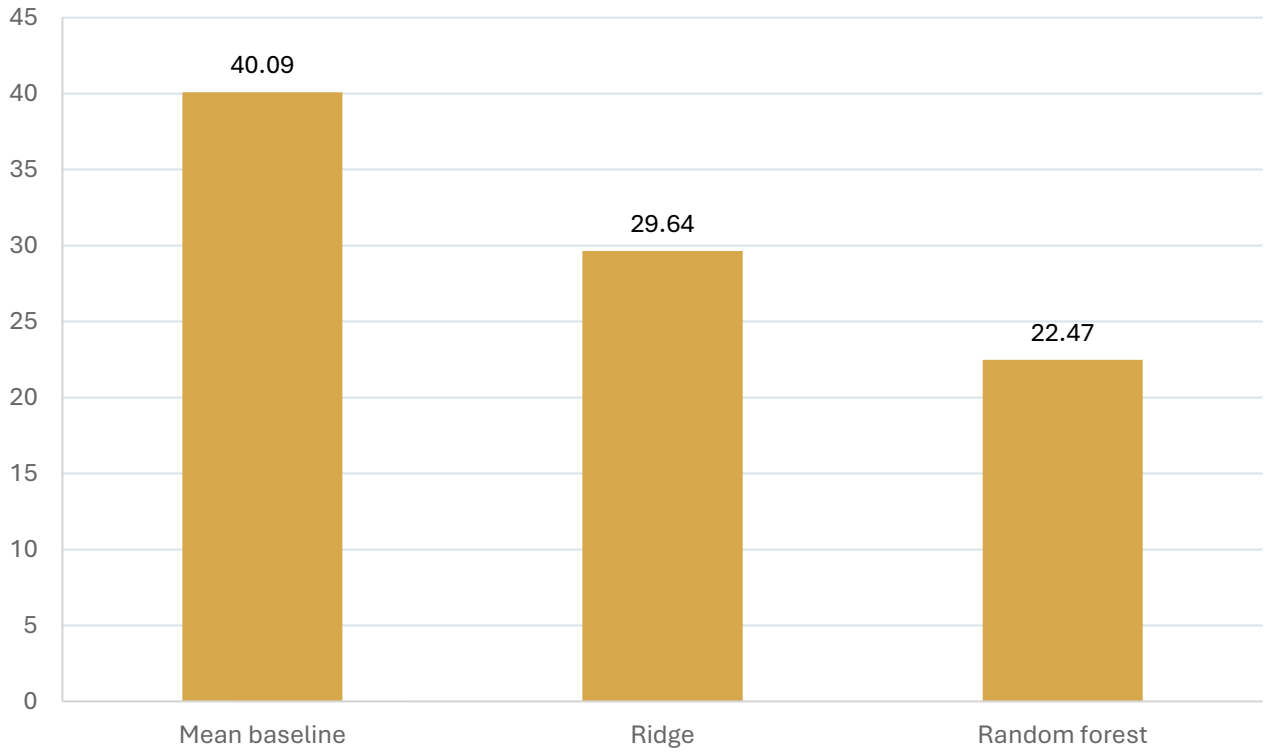
79.8%

precision among contacted bookings

Random forest scores slightly higher on the test period, but test results did not change the model-selection rule.

TASK 6

ADR model cuts test MAE by 43.9%



Selected model

Random forest

€22.47

average absolute test error

0.649

test R^2

Hotel, arrival timing and market segment are the strongest measured price drivers.

TASK 7

Two decision rules connect predictions to work

Cancellation review

10,300

bookings contacted

52.3%

cancellations captured

Action: send a confirmation reminder. Staff review contract and deposit eligibility before any request.

2.06× lift over random selection at the same contact volume

ADR review

1,465

bookings flagged

€58,839

gross review opportunity

Action: route quotes more than €20 below the model expectation to revenue management before confirmation.

Opportunity value is a review estimate, not guaranteed revenue

TASK 8

Recommendations and deployment controls

Start with a monitored pilot

Use cancellation reminders for the highest-risk quartile. Route ADR gaps to revenue management. Record staff decisions and outcomes so the next model learns from actual interventions.

Monthly controls

Monitor PR-AUC, precision, recall, ADR MAE and bias by hotel and segment. Retrain when performance crosses the documented thresholds for two review periods.

Known limitations

No booking ID or room-capacity history. Revenue excludes extras, taxes and refunds. Model actions require human review.

AI supported drafting and leakage review. The team reran every notebook and checked the reported numbers.